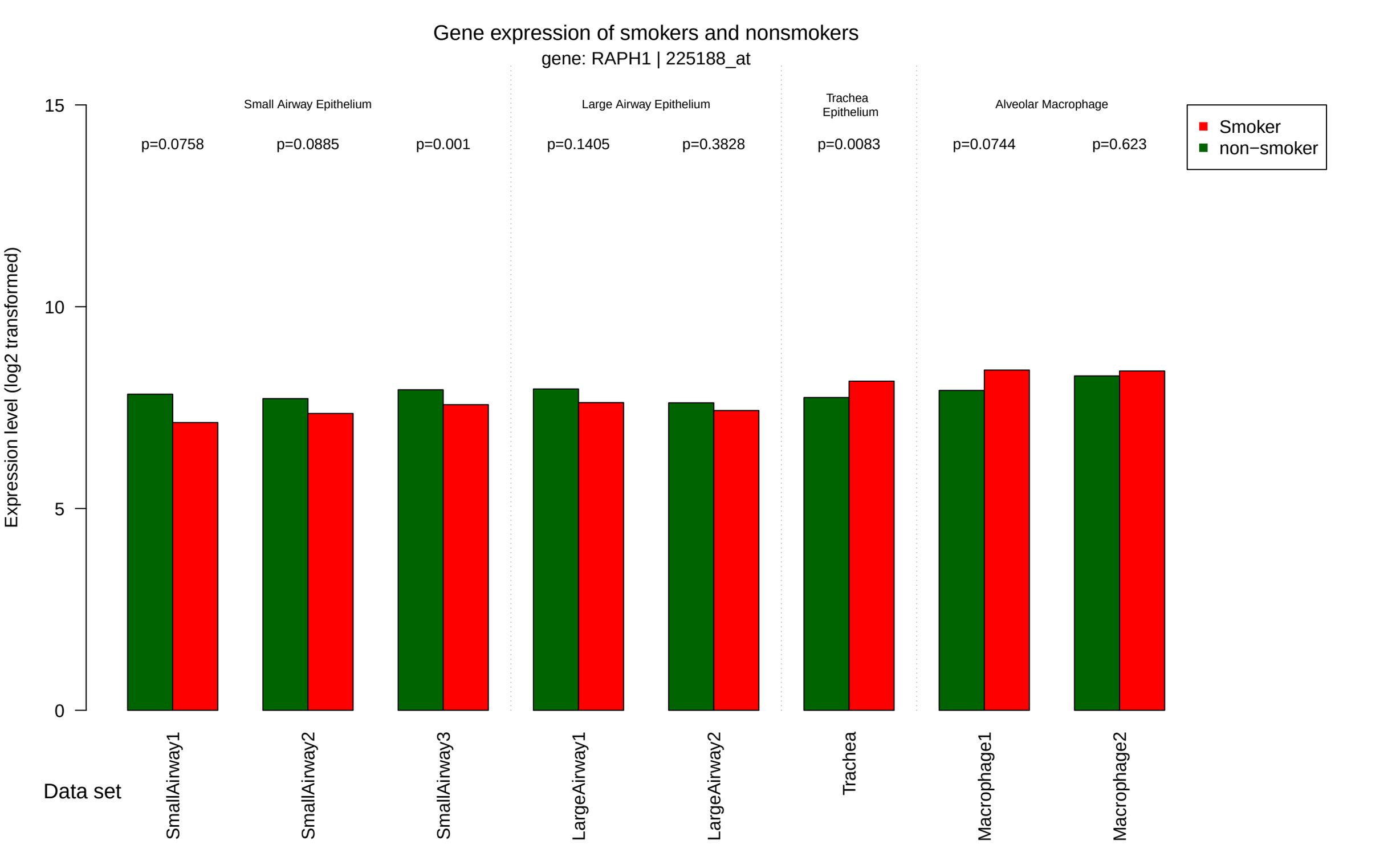
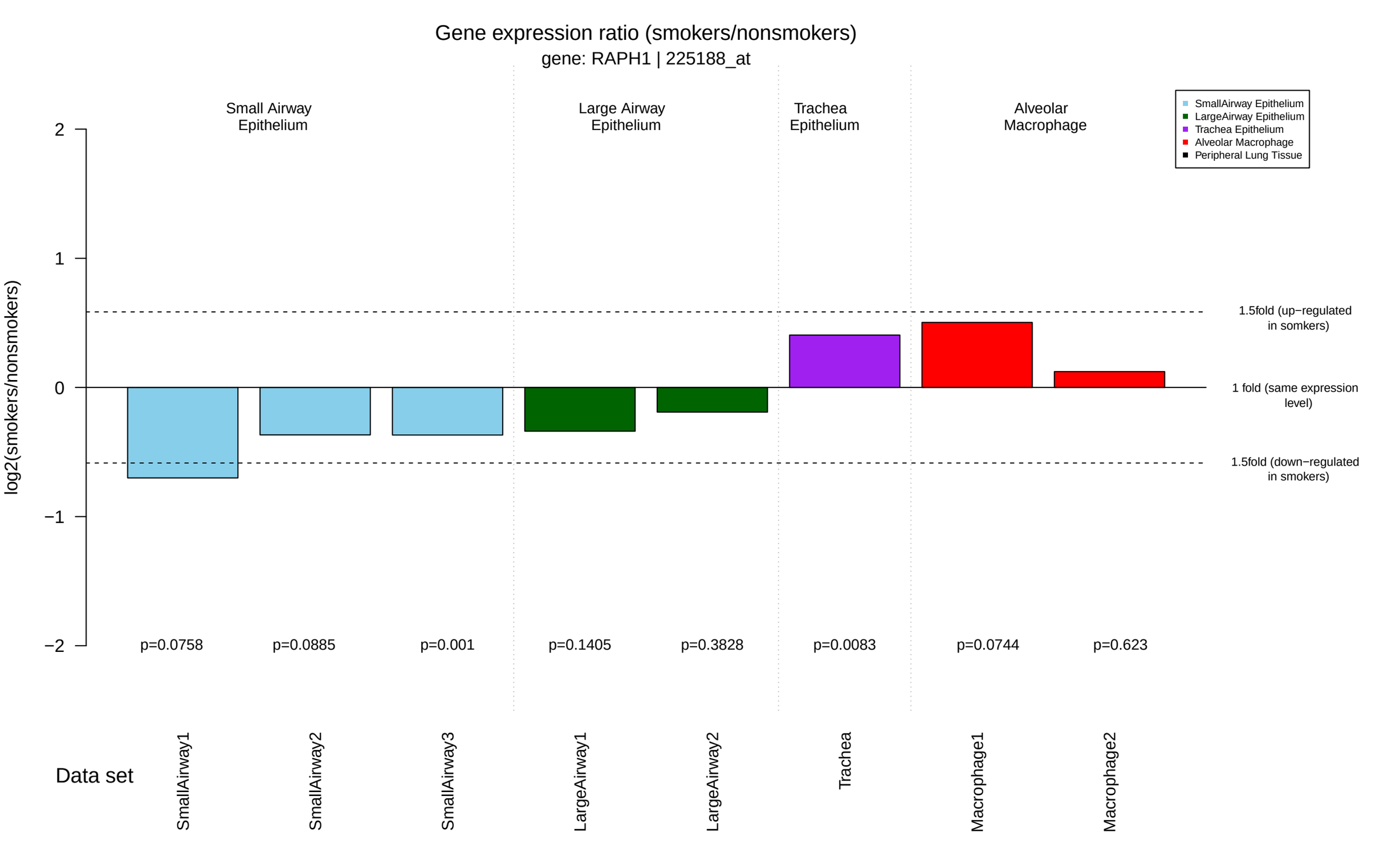
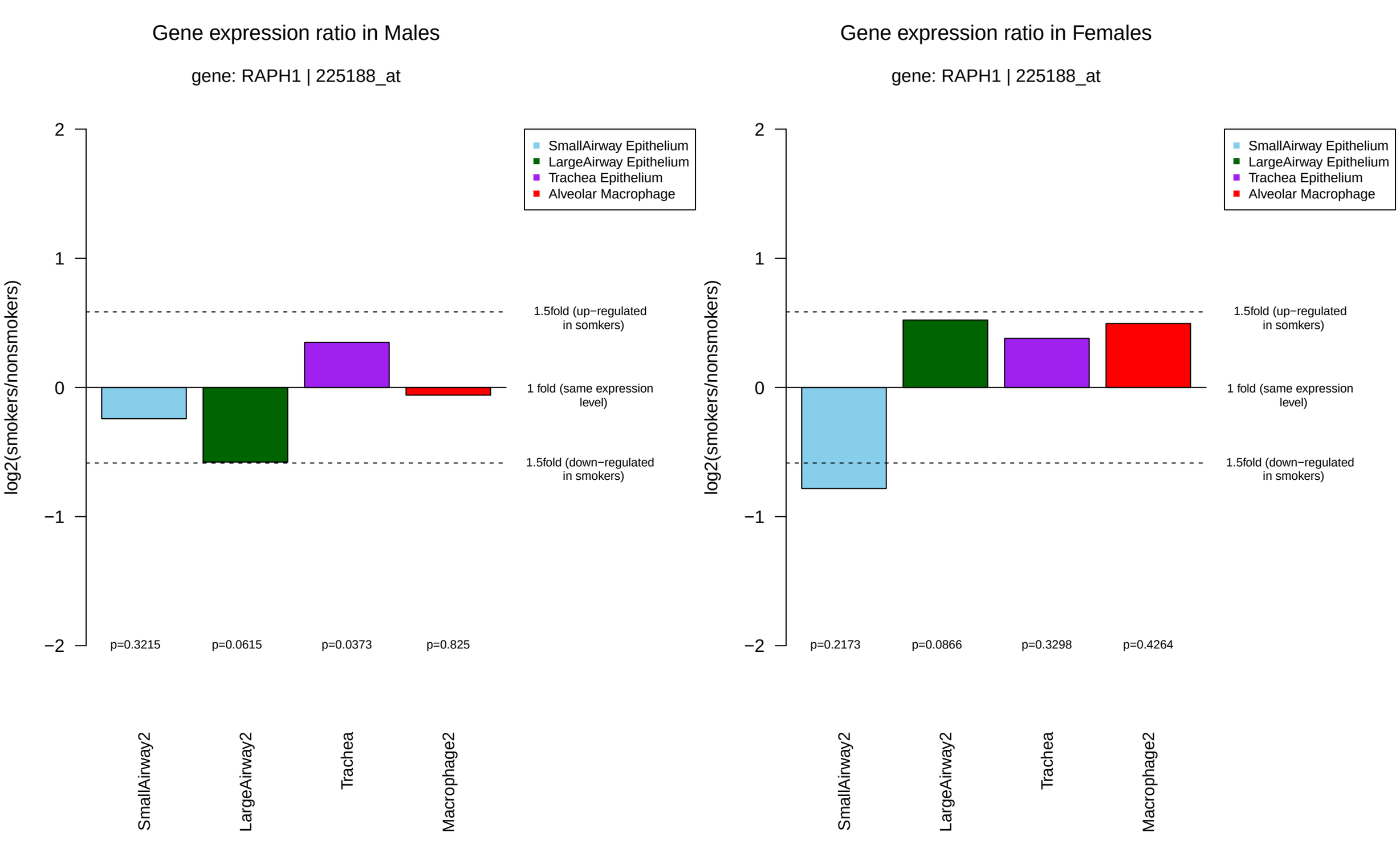
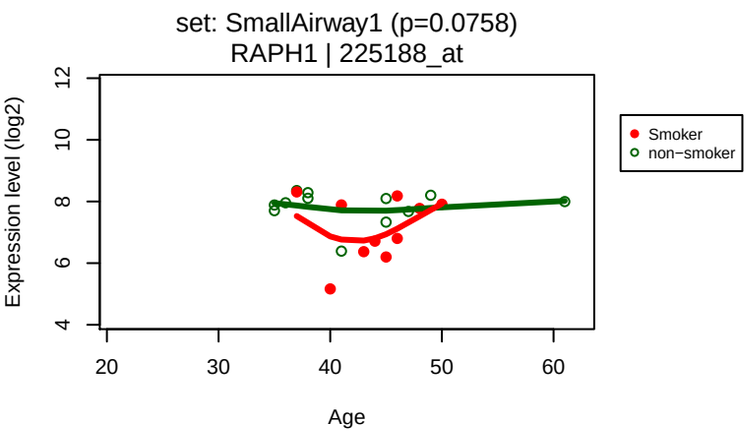
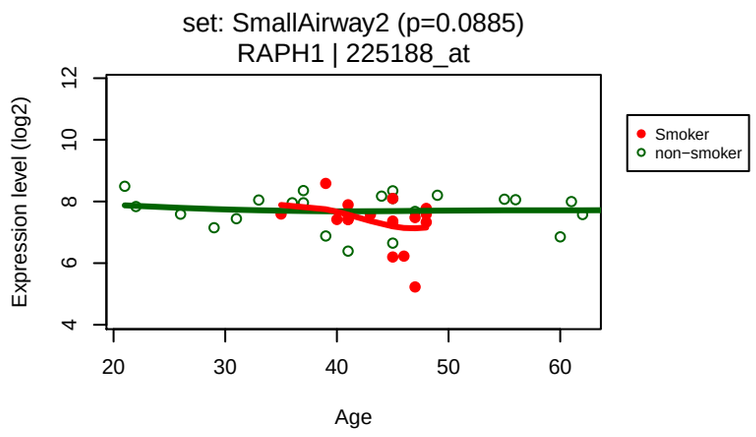
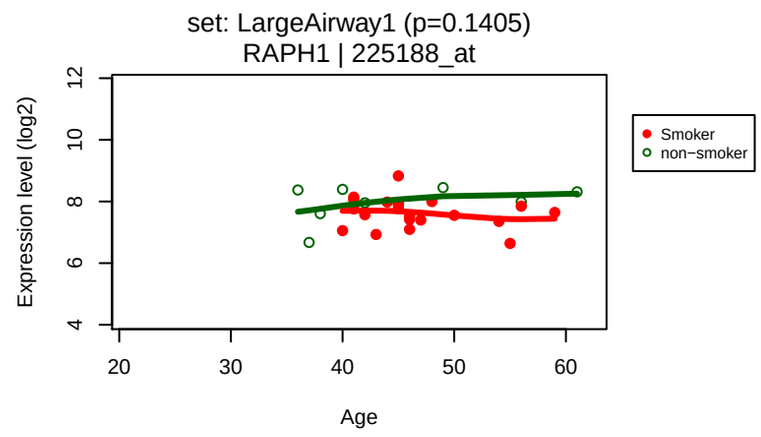

Gene expression of smokers and nonsmokers
gene: RAPH1 | 225188_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0758
p=0.0885
p=0.001
p=0.1405
p=0.3828
p=0.0083
p=0.0744
p=0.623
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

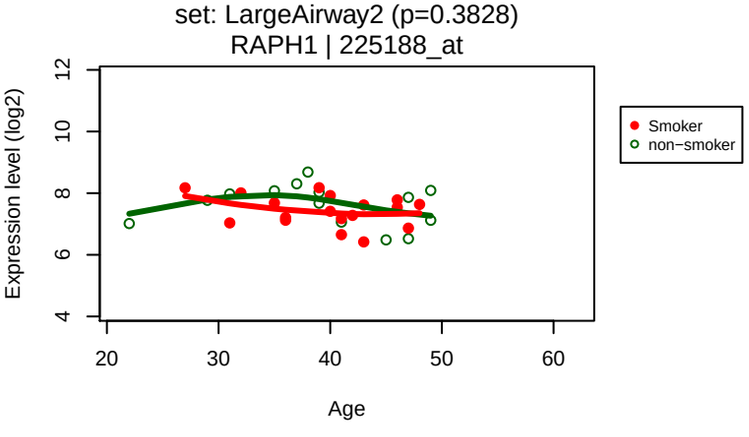

Gene expression ratio (smokers/nonsmokers)
gene: RAPH1 | 225188_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0758
p=0.0885
p=0.001
p=0.1405
p=0.3828
p=0.0083
p=0.0744
p=0.623
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

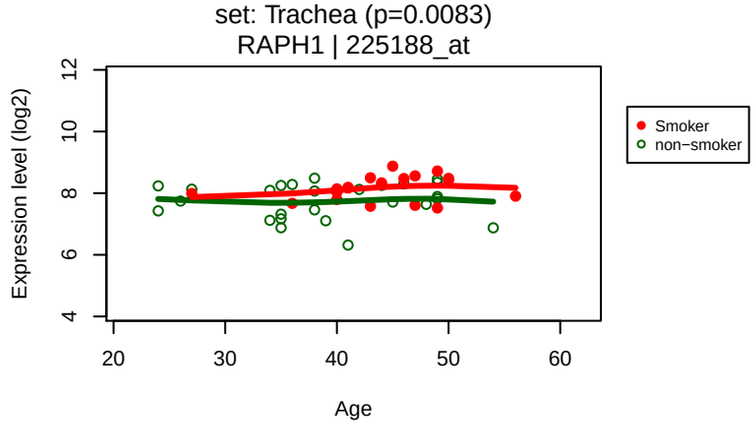

Gene expression ratio in Males
gene: RAPH1 | 225188_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3215
p=0.0615
p=0.0373
p=0.825
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAPH1 | 225188_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2173
p=0.0866
p=0.3298
p=0.4264
SmallAirway2
LargeAirway2
Trachea
Macrophage2

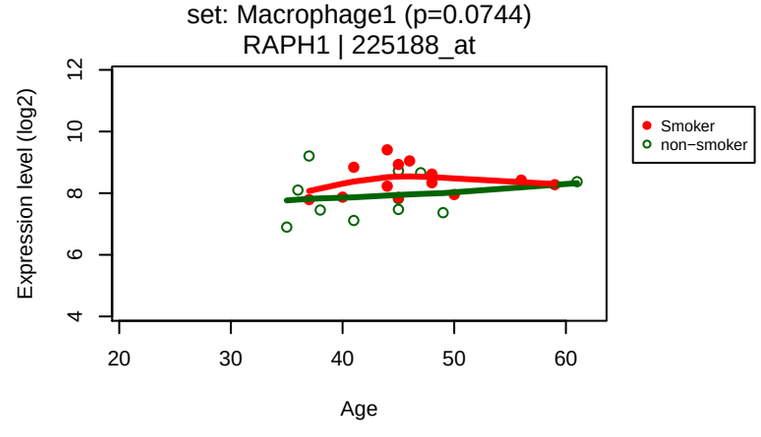

set: SmallAirway1 (p=0.0758)
RAPH1 | 225188_at
Expression level (log2)
Age
Smoker
non−smoker

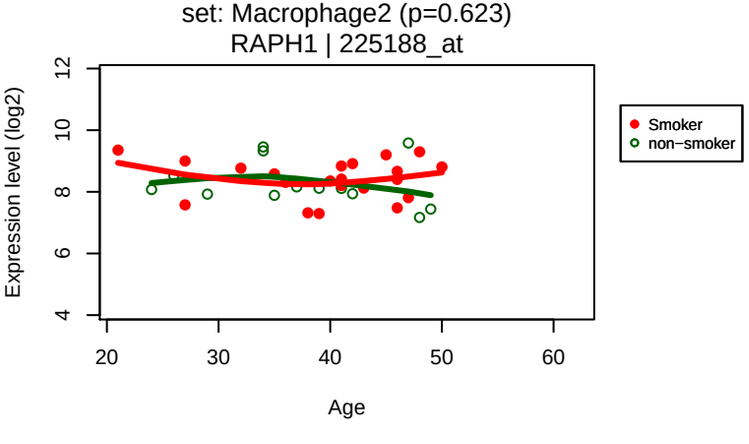

set: SmallAirway2 (p=0.0885)
RAPH1 | 225188_at
Expression level (log2)
Age
Smoker
non−smoker

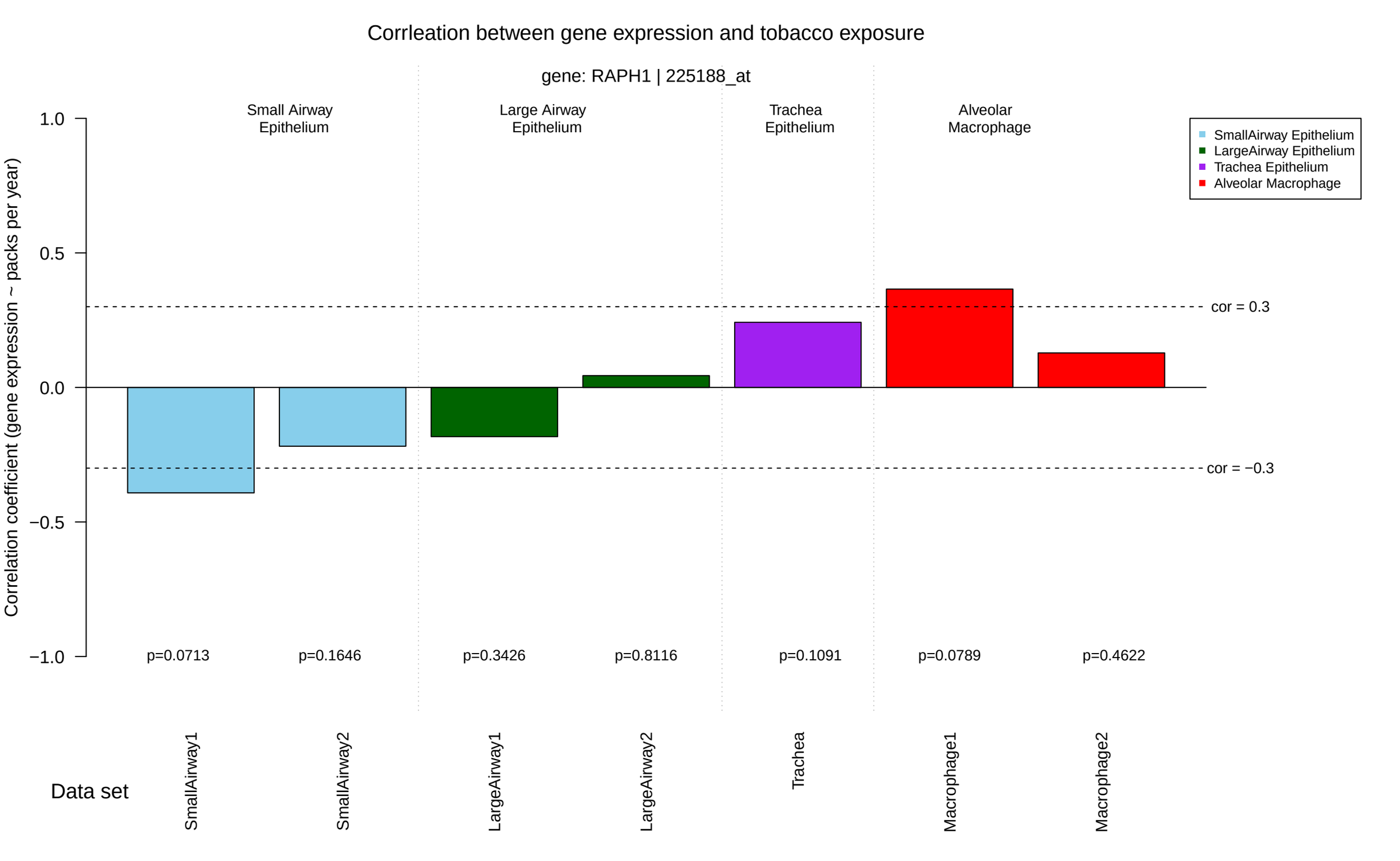

set: LargeAirway1 (p=0.1405)
RAPH1 | 225188_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3828)
RAPH1 | 225188_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0083)
RAPH1 | 225188_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0744)
RAPH1 | 225188_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.623)
RAPH1 | 225188_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAPH1 | 225188_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0713
p=0.1646
p=0.3426
p=0.8116
p=0.1091
p=0.0789
p=0.4622
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2